



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknološi ya Tshedimošo



Unit 11: Setting the MARR

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh

Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- The significance of cost and revenue projection.
- Some significant cost components commonly used in cost and revenue projection.
- Ability to use the “bottom-up” or “top-bottom” business system operational activities for cost and revenue projection.
- Ability to carry out cost and revenue projection into the future.
- Ability to estimate Best, Optimal and Worst Case budget scenarios.
- How all of these concepts can be applied to the semester project.

Learning Outcomes for the Current Lecture

At the end of this lecture you should be able to:

- **Discuss**, **analyse**, **apply**, and **evaluate** the MARR, including its driving factors
- **Discuss**, **analyse**, **apply**, and **evaluate** the structure and source of capital
- **Distinguish** between debt and equity financing
- **Calculate** the weighted average cost of capital
- Set the MARR for your project by **creating** your own financial scenario and applying the above discussion
- **Explain**, **apply**, and **evaluate** the outcomes of interest rates

Presentation Outline

- Explain the concept of MARR in the context of an investment
- Discuss factors that influence the setting of the MARR.
- Capital Structure and Sourcing
- Differentiate between debt and equity financing.
- Calculate the weighted average cost of capital.
- Discuss and apply other forms of interest rates
- Apply the entire knowledge to the semester project.

What is MARR in a simple context from the point of view of an investor?

MARR: Minimum Acceptable Rate of Return

Minimum Attractive Rate of Return

Also called the *Hurdle Rate*



"MARR, or hurdle rate, is the minimum rate of return on a project a manager or company is willing to accept before starting a project, given its risk and the opportunity cost of forgoing other projects."

- Wikipedia 😊

MARR: Minimum Acceptable Rate of Return Minimum Attractive Rate of Return



project cash flow

MARR

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If the project cash flow can jump over the hurdle, it is said to be financially feasible.

1. How do you determine how high the hurdle should be? (Unit 11)
2. How do you determine whether the project cash flow jumped over the hurdle? (Unit 12)

**Discuss factors that influence setting of the
MARR**

Factors that influence the setting of the MARR

- The weighted average cost of capital (WACC) $WACC_{debt}$, $WACC_{equity}$
- Cost of the project (project budget)
- Capital sources and the cost of each type of capital
- Risks and other debilitating factors that can impact a business
- Market rate (benchmarking)
- Planning horizon

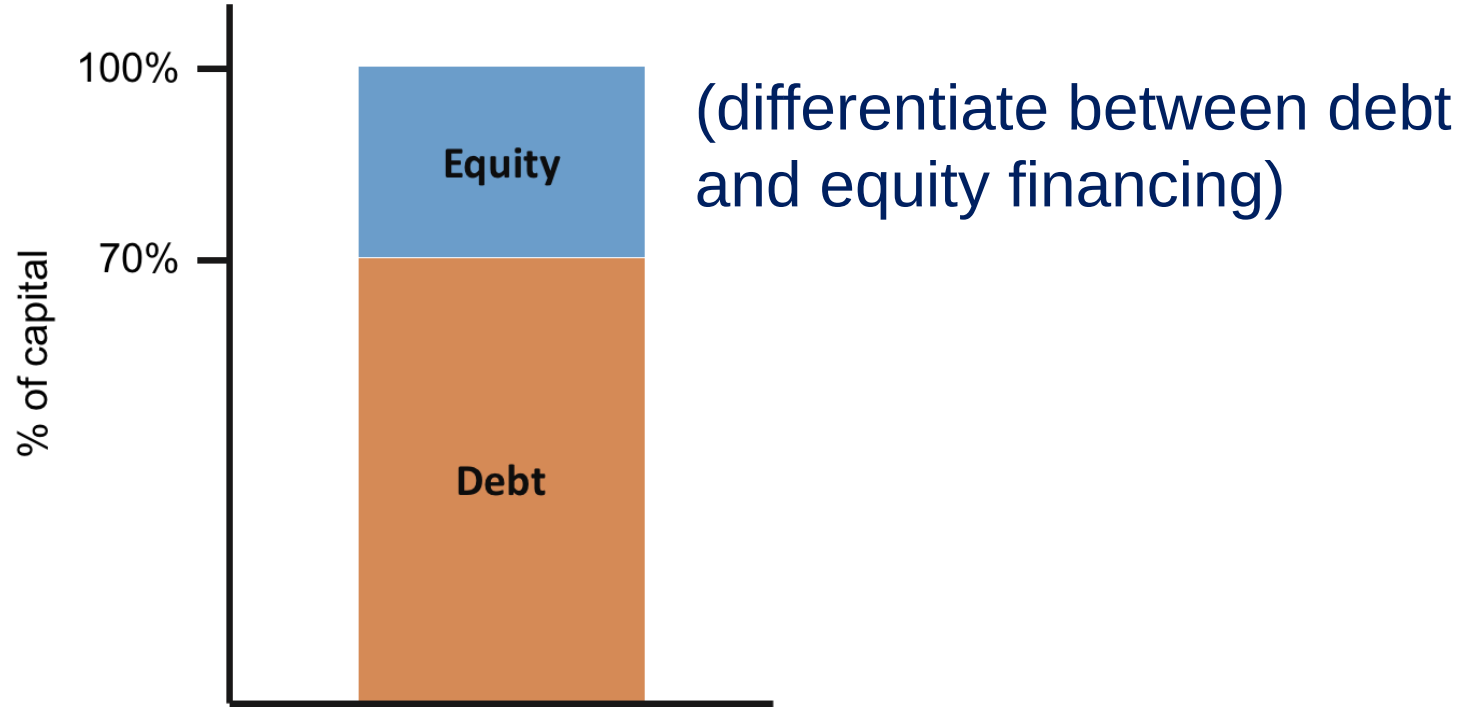
Factors that influence the setting of the MARR..... ...Contd.

- Management favour
- Corporate tax
- Limited capital

Structure and Source of Capital

MARR is often “set” on capital investments hence, a need to explore a few issues around the **structure** and **sourcing** of capital.

Capital structure



Capital structure

Debt = outsiders' money

Loans or mortgages

Bonds

- Payment terms are set. No claim in profits.
- Can be riskier than equity.

What is a bond?:

<https://www.youtube.com/watch?v=luyejHOGCro>

Equity = owner's money

Shares (*Stocks U.S.A*)

- Diluting ownership in exchange for capital
- Share in profits but no claim in case of losses

Retained earnings

Bond tangent...

For your interest only:

<https://www.moneyweb.co.za/investing/the-downgrade-and-your-investments/>

<https://www.businesslive.co.za/bd/markets/2020-04-02-foreign-investors-dropped-sa-bonds-ahead-of-downgrade/>

<https://www.businessinsider.co.za/government-bonds-south-africa-2020-3>

How to calculate the “WACC”

(weighted average cost of capital)

How do you calculate your semester mark?

Assessment	Weight	Your mark
Class preparation and participation exercises	5%	100%
Group project (PART A)	50%	67%
Group project (PART B)	35%	75%
Exam preparation tutorials	10%	94%

$$\text{Semester mark} = 0.05(100\%) + 0.5(67\%) + 0.35(75\%) + 0.1(94\%) = 74.15\%$$

The Analogy on the previous slide is the same with the WACC calculation

Capital source	Weight	Cost of capital
Loan (debt)	30%	14%
Corporate bonds issued (debt)	25%	11%
Shares issued (equity)	15%	9.5%
Retained earnings (equity)	30%	6%

$$\text{WACC} = 0.3(14\%) + 0.25(11\%) + 0.15(9.5\%) + 0.3(6\%) = 10.18\%$$

WACC, $WACC_{\text{equity}}$, $WACC_{\text{debt}}$

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Loan (debt)	30%	14%
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Shares issued (equity)	15%	9.5%
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$$WACC_{\text{debt}} = \frac{0.30}{0.55}(14\%) + \frac{0.25}{0.55}(11\%)$$

$$WACC_{\text{debt}} = 7.64\% + 5\% = 12.64\%$$

WACC, $WACC_{\text{equity}}$, $WACC_{\text{debt}}$

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$$WACC_{\text{equity}} = \frac{?}{?}(9.5\%) + \frac{?}{?}(6\%)$$

$$WACC_{\text{equity}} = ? + ? = 7.17\%$$

WACC, $WACC_{\text{equity}}$, $WACC_{\text{debt}}$

$$\mathbf{WACC} = \text{weight of equity}(WACC_{\text{equity}}) + \text{weight of debt}(WACC_{\text{debt}})$$

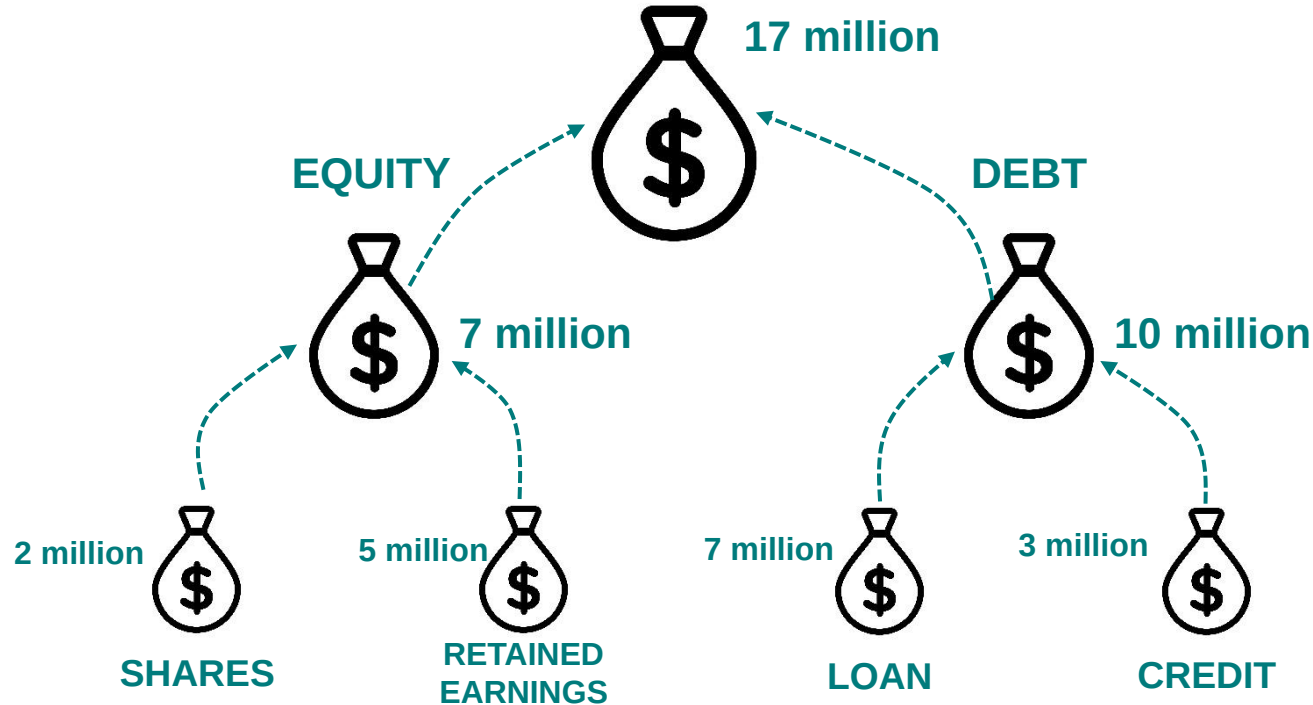
$$\mathbf{WACC} = 0.45(7.17\%) + 0.55(12.64\%) = 10.8\%$$

Example: Nancy Botwin Medicinal Distillers

Nancy Botwin Medicinal Distillers need to finance the expansion of one of their product lines. The expansion project will require a capital investment of R17 million. They will raise R2 million through shares where shareholders receive a return on investment of 11%. Further they will use R5 million of retained earnings currently in a money market account where it earns 5%. A further R7 million will be financed through a mortgage loan secured by new facilities. An interest rate of 11.5% applies to the mortgage. The remainder will be financed using their existing credit facility at the bank (interest rate 10.5%). All interest rates are compounded per annum.

1. What is the $WACC_{\text{equity}}$?
2. What is the $WACC_{\text{debt}}$?
3. What is the WACC?

Example: Nancy Botwin Medicinal Distillers



Example: Nancy Botwin Medicinal Distillers

$$\text{WACC}_{\text{equity}} = \frac{2}{7}(\text{cost of shares}) + \frac{5}{7}(\text{cost of RE})$$

$$\text{WACC}_{\text{debt}} = \frac{7}{10}(\text{cost of loan}) + \frac{3}{10}(\text{cost of credit})$$

$$\text{WACC} = \frac{7}{17}(\text{WACC}_{\text{equity}}) + \frac{10}{17}(\text{WACC}_{\text{debt}})$$

Example: Nancy Botwin Medicinal Distillers

$$\text{WACC}_{\text{equity}} = \frac{2}{7}(11\%) + \frac{5}{7}(5\%) = 6.71\%$$

$$\text{WACC}_{\text{debt}} = \frac{7}{10}(11.5\%) + \frac{3}{10}(10.5\%) = 11.2\%$$

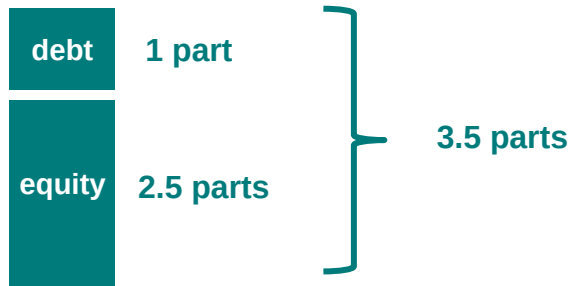
$$\text{WACC} = \frac{7}{17}(6.71\%) + \frac{10}{17}(11.2\%) = 9.35\%$$

Example: Weintraub Electronics

Weintraub Electronics wants to expand to a new market segment with their ActiveWear wireless earphones. The cost of the R3 million they raised in equity capital is 13% per year. Their debt-equity ratio for this project is 0.4, (which can also be interpreted as debt:equity of 1:2.5). Half of their debt financing is from a loan at an interest rate of 11% per year while the rest is loaned from a stakeholder at 14% per year. What is the WACC for this project?

Example: Weintraub Electronics

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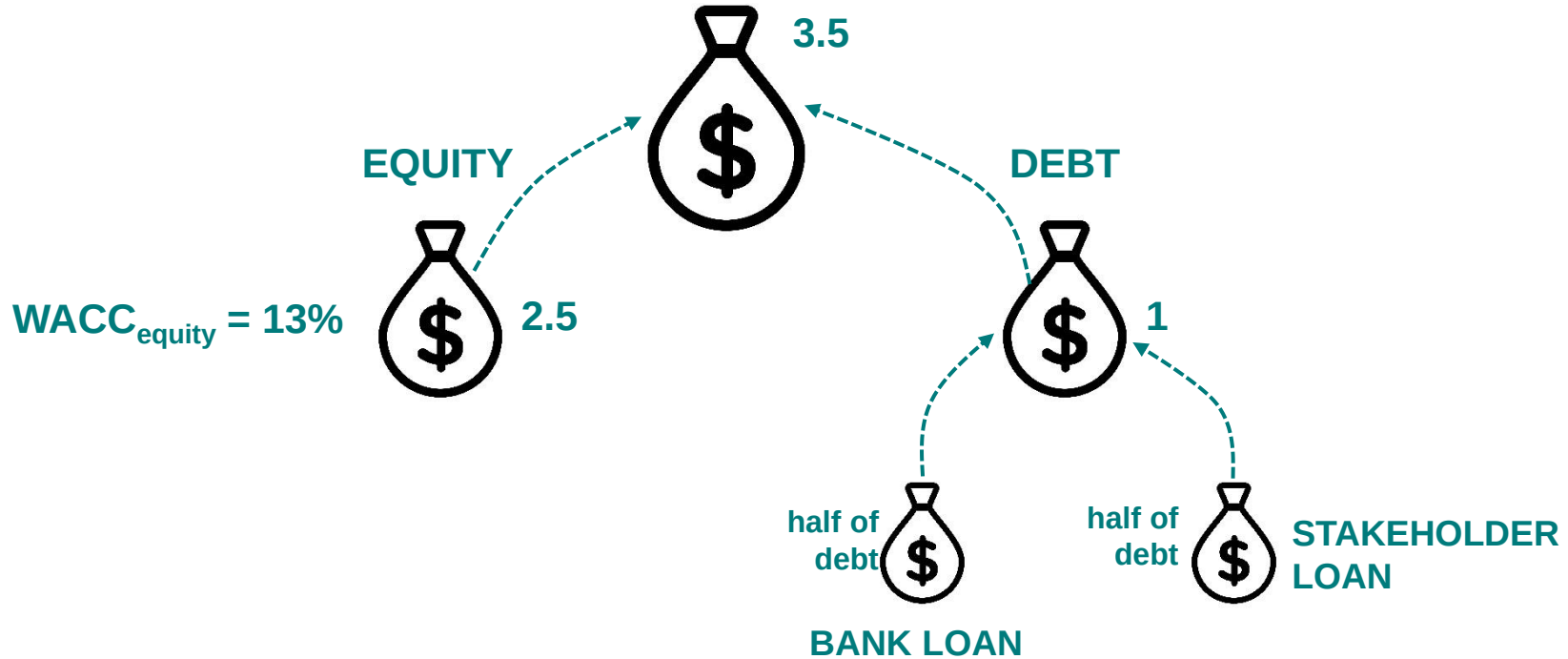


$$\text{weight of debt} = \frac{1}{3.5} = R ?$$

$$\text{weight of equity} = \frac{2.5}{3.5} = R ?$$



Example: Weintraub Electronics



Example: Weintraub Electronics

$$\text{WACC}_{\text{equity}} = 13\%$$

$$\text{WACC}_{\text{debt}} = 0.5(11\%) + 0.5(14\%) = 12.5\%$$

$$\text{WACC} = \frac{2.5}{3.5}(13\%) + \frac{1}{3.5}(12.5\%) = 12.86\%$$

or...

$$\text{WACC} = \frac{2.5}{3.5}(13\%) + \frac{1}{3.5}[0.5(11\%) + 0.5(14\%)] = 12.86\%$$

Example: Weintraub Electronics (Alternative Approach)

Debt : Equity = (1 : 2,5)

Debt : Equity = (x : 3million) = (1,2million : 3million)

Equity	RE	3million	13%
	Shares		
	Total weight Equity	3million	
Debt	Loan-1	0,6million	11%
	Loan-2	0,6million	14%
	Total weight Debt	1,2million	
Grand Total		4,2million	

Example: Weintraub Electronics (Alternative Approach)

$$\text{WACC}_{\text{Equity}}: \frac{3}{3} * (13\%) =$$

13%.....Equity

$$\text{WACC}_{\text{Debt}}: \frac{0,6}{1,2} * (11\%) + \frac{0,6}{1,2} * (14\%) = 5,5 + 7 =$$

12,5%.....Debt

$$\text{WACC}_{\text{Total}}: \frac{3}{4,2} * (13\%) + \frac{1,2}{4,2} * (12,5\%) = 9,2857 + 3,5714 \approx 12,86\%$$

Setting the MARR for your project

Other Interest Rates

Repo rate: the rate at which the South African Reserve Bank for instance, lends money to the banks. 8.25%

Prime lending rate - the rate at which the banks lend to their customers 11.75%

Reverse Repo Rate-This is the rate the Reserve or Central bank of a nation pays its commercial banks to park their excess funds in the central bank. Reverse repo rate is also a monetary policy used by the central bank to regulate the flow of money in the market.

Other Interest Rates

Inflation rate

South Africa's inflation rate 2024

5.3% in January

5.6% in February

5.5% market forecasts

4.5% central bank's preferred midpoint of the 3–6% target range.

Inflation index driver

$$\text{Inflation Formula} = \frac{(\text{CPI}_{x+1} - \text{CPI}_x)}{\text{CPI}_x}$$

CPI= Consumer price index

Inflation Rate Driver

Given table 3, a basket of selected food items, you are required to use Dec. 2022 as a base year hence, required to determine if there has been an increase in inflation between Dec. 2023 and Mar. 2024.

Inflation Rate Driver

Goods	Dec-2022 R/unit	Dec-2023 R/unit	Mar-2024 R/unit
Baked beans – tinned 410g	13,85	15,87	16,05
Instant coffee 250g	48,17	58,55	60,34
Eggs 1.5 dozen	54,84	68,49	68,52
Sunflower oil 750ml	38,71	35,82	35,18
Apples per kg	19,84	23,76	23,41
Oranges per kg	18,84	24,20	38,77
Beef mince per kg	106,62	104,37	103,49
Polony 1kg	51,44	54,58	55,45
Rice 2kg	35,82	43,19	46,17

**Table: Basket of goods for
selected food items**

**Source: Statssa: NAMC
FOOD BASKET (2022,
2023 & 2024)**

Inflation Rate Driver....Contd.

Goods	Dec-2022 R/unit	Dec-2023 R/unit	Mar-2024 R/unit
Loaf of white bread 700g	18,62	18,59	18,43
White sugar 2,5g	50,27	61.64	62,88

**Table Contd...: Basket of
goods for selected food items**

**Source: Statssa: NAMC FOOD
BASKET (2022, 2023 & 2024)**

Inflation Rate Driver....Contd.

$$\Sigma \text{prices for Dec. 2022} = 457,02$$

$$\Sigma \text{prices for Dec. 2023} = 509,06$$

$$\Sigma \text{prices for Mar. 2024} = 528,69$$

$$CPI = \frac{\text{Price of basket of goods in a year}}{\text{Price of basket of goods in a base year}} * 100$$

Inflation Rate Driver....Contd.

$$CPI_{Dec.2022} = \frac{457,02}{457,02} * 100 = 100$$

$$CPI_{Dec.2023} = \frac{509,06}{457,02} * 100 \approx 111,387$$

$$CPI_{Mar.2024} = \frac{528,69}{457,02} * 100 \approx 115,682$$

Inflation Rate Driver....Contd.

Inflation: 2022 – 2023

$$= \frac{CPI_{Dec.2023} - CPI_{Dec.2022}}{CPI_{Dec.2022}} * 100$$

$$= \frac{111,387 - 100}{100} * 100$$

$$\approx 11,387\%$$

Inflation Rate Driver....Contd.

Inflation: 2022 – 2024

$$= \frac{CPI_{Mar.2024} - CPI_{Dec.2022}}{CPI_{Dec.2022}} * 100$$

$$= \frac{115,682 - 100}{100} * 100$$

$$\approx 15,682$$

The above indicators show that there has been a growth in inflation between: (i) Dec. 2022 and Dec. 2023; (ii) Dec. 2022 and Mar. 2024; and (iii) Dec.2023 and Mar.2024 (though 2023 was not used as a base year)

Monthly Loan Repayment

$$\text{Monthly Payment (PMT)} = \frac{P \left[\frac{r}{n} \right]}{1 - \left(1 + \frac{r}{n} \right)^{-nt}}$$

$$\text{Initial Interest Payment } i = P * \frac{r}{n}$$

Where r = interest rate per annum

n = number of payments per year

t = duration of investment

P = principal

i = interest paid per monthly period

$$\text{Initial Interest Payment } i = P * \frac{r}{12}$$

Monthly Payment Remarks

- The payment returned by PMT includes principal and interest but no taxes, reserve payments, or fees sometimes associated with loans.
- Make sure that you are consistent about the units you use for specifying **rate** and **duration**.

For instance, if you make monthly payments on a four-year loan at an annual interest rate of 12 percent, use $(12\%/12)$ for rate and $(4*12)$ for duration. If you make annual payments on the same loan, use 12% for rate and 4 for duration.

Microsoft Excel Syntax for Monthly Payment

Syntax **PMT**(rate, nper, pv, [fv], [type])

Rate (Required) The interest rate for the loan. For monthly payments, divide this by 12.

Nper (Required) The total number of payments for the loan. For monthly payments, multiply this by 12.

Pv (Required) The present value, or the total amount that a series of future payments is worth now; also known as the principal. In other words, the loan amount.

Monthly Payment Remarks

Fv (Optional) The future value, or the balance you want after the last payment is made. If you don't include a value for **fv**, it's assumed to be 0 (zero), that is, the future value of the loan is 0.

Type (Optional) The number 0 (zero) means the payments are due at the end of each period, and 1 means they're due at the beginning.

Monthly Payment Example

A BSS310 student was able to secure a 30-year R400000 property loan to start up a real estate business. Assuming the interest rate was fixed at 5%, you are required to address the following:

- (a) Calculate the monthly mortgage payment.
- (b) How much would be paid in the 1st monthly payment to the bank as interest?
- (c) Determine both the capital and interest payments in the 3rd month.

Monthly Payment Remarks

$$\text{Monthly Payment (PMT)} = \frac{P\left[\frac{r}{n}\right]}{1 - \left(1 + \frac{r}{n}\right)^{-nt}}$$

$$= \frac{400000 * \left[\frac{5\%}{12}\right]}{1 - \left(1 + \frac{5\%}{12}\right)^{-12*30}} = R2147,29$$

Monthly Payment (Amortization Table)

By using interest payment $(i) = P * \frac{r}{12}$, the table below can be obtained

Payment Sequence	Date	Principal	Interest	Total Interest	Balance
Initialised Scenario		-----	-----	-----	R400000
First Payment		R480,62	R1666,67	R1666,67	R399519,38
Second Payment		R482,63	R1664,66	R3331,33	R399036,75
Third Payment		R484,64	R1662,65	R4993,98	R398552,11

Learning Outcomes Evaluated

- Are you able to discuss, analyse, apply, and evaluate the MARR, including its driving factors? Can you extend this knowledge to the semester project?
- Are you able to discuss, analyse, and evaluate the structure and source of capital? Can you extend this knowledge to the semester project?
- Can you distinguish between debt and equity financing
- Can you calculate the weighted average cost of capital
- Can you set the MARR for the semester project by creating your own financial scenario and applying the above discussion
- Can you explain some vital interest rates and apply them as deemed necessary?

The End

Next Topic

- Unit 12: Time value of money



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